

Homework 6

● Graded

Student

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Total Points

38 / 38 pts

Question 1

Problem 1

6 / 6 pts

✓ - 0 pts Correct

- 1 pt (1.1) $X | (Y=0) \sim \text{Bern}(p)$ or $E(X | Y=0) = \frac{1}{3}$

- 1 pt (1.1) $p = P(X=1 | Y=0) = \frac{1}{3}$ or $E(X^2 | Y=0) = \frac{1}{3}$

- 1 pt (1.1) $\text{Var}(X | Y=0) = \frac{2}{9}$

Click here to replace this description.

- 1 pt (1.2) $P(X=1, A) = \frac{5}{16}$

- 1 pt (1.2) $P(A) = \frac{13}{16}$

- 1 pt (1.2) $P(X=1 | A) = \frac{5}{13}$

- 6 pts Incorrect

Question 2

Problem 2

8 / 8 pts

✓ - 0 pts Correct

- 1 pt part a: error in integrating when finding $E[X]$ - 1 pt part a: error in integrating when finding $E[Y]$ - 2 pts part a: missing $E[Y]$ calculation- 1 pt part b: error in integrating with $\text{Cov}(X, Y)$

- 1 pt part b: calculation error

- 2 pts part b: error in covariance formula

- 3 pts part b: missing

- 0.5 pts reasoning in c incorrect

note that covariance being 0 does not imply independence. the fact that $f_{X,Y}(x, y) = f_X x f_Y y$ is why we can say these are independent random variables and is an if and only if/bidirectional truth. A covariance of 0 does not imply two variables are independent, but if two random variables are independent, then their covariance will be 0 and thus they will be uncorrelated.

- 0.5 pts missing reasoning in c

- 1 pt part c: missing

- 8 pts missing

Question 3

Problem 3

8 / 8 pts

✓ - 0 pts Correct

- 2 pts (a) probability that at least one of Sarah and Hannah caught typo is $p = P(\text{Sarah caught typo} \cup \text{Hannah caught typo}) = P(\text{Sarah caught typo}) + P(\text{Hannah caught typo}) - P(\text{both Sarah and Hannah caught typo}) = p_1 + p_2 - p_1 p_2$ (or can be calculated using $1 - (1 - p_1)(1 - p_2) = p_1 + p_2 - p_1 p_2$): incorrect probability

- 2 pts (a) Let X be the number of typos found by at least one proof readers, $X \sim \text{Binom}(n, p_1 + p_2 - p_1 p_2)$: incorrect distribution

- 2 pts (b) $X_1 + X_2$ is sum of $2n$ independent Bernoulli trials where the probability of success is p , i.e., $X_1 + X_2 \sim \text{Binom}(2n, p)$: incorrect $X_1 + X_2$ distribution

- 2 pts (b) conditional distribution pdf $P(X_1 = k | X_1 + X_2 = t)$ is $\frac{P(X_1=k, X_1+X_2=t)}{P(X_1+X_2=t)} = \frac{P(X_1=k, X_2=t-k)}{P(X_1+X_2=t)} = \frac{P(X_1=k)P(X_2=t-k)}{P(X_1+X_2=t)} = \frac{\binom{n}{k} p^k (1-p)^{n-k} \binom{n}{t-k} p^{t-k} (1-p)^{n-(t-k)}}{\binom{2n}{t} p^t (1-p)^{2n-t}} = \frac{\binom{n}{k} \binom{n}{t-k}}{\binom{2n}{t}}$, which is Hypergeometric($2n, n, p$): incorrect conditional distribution pdf

- 8 pts wrong submission

- 3 pts (b) Missing derivation process

Question 4

8 / 8 pts

Problem 4

✓ - 0 pts Correct

- 0.5 pts space incorrect in part a. Nonzero probability should be in $k = 2, 3, \dots, 10$
- 1 pt pmf incorrect in part a
- 0.5 pts space incorrect in part b. Nonzero probability should be in $k = 2, 3, \dots, 10$
- 1 pt pmf incorrect in part b
- 1 pt need to state explicitly what $P(X=0)$ and $P(X=1)$ is if you have not specified the distribution for X
- 1 pt pmf not specified as a piecewise function
- 4 pts part b missing
- 8 pts problem missing
- 3 pts used cdf here for both parts

Question 5

4 / 4 pts

Problem 5

✓ - 0 pts Correct

- 1 pt $P(X + Y = k) = \sum_{i=0}^k P(X = i, Y = k - i)$: incorrect setup
- 1 pt $P(X + Y = k) = \sum_{i=0}^k P(X = i)P(Y = k - i) = \sum_{i=0}^k e^{-\lambda_1 - \lambda_2} \frac{\lambda_1^i}{i!} \frac{\lambda_2^{k-i}}{(k-i)!}$: incorrect use of independence and pdf
- 1 pt $P(X + Y = k) = \sum_{i=0}^k e^{-\lambda_1 - \lambda_2} \frac{\lambda_1^i}{i!} \frac{\lambda_2^{k-i}}{(k-i)!} = \frac{e^{-\lambda_1 - \lambda_2}}{k!} \sum_{i=0}^k k \frac{k!}{i!(k-i)!} \lambda_1^i \lambda_2^{k-i}$: incorrect/incomplete derivation
- 1 pt recognizing that $P(X + Y = k) = \frac{(\lambda_1 + \lambda_2)^k}{k!} e^{-(\lambda_1 + \lambda_2)}$ means $X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$: failure to recognize the distribution or incorrect pdf
- 3 pts missing derivation
- 4 pts wrong submission
- 4 pts missing solution

Question 6

4 / 4 pts

Problem 6

- 1 pt $Var(X) = \sum_{i=1}^n Var(X_i) + \sum_{i \neq j} Cov(X_i, X_j)$, or $Var(X) = \sum_{i=1}^n E(X_i^2) + \sum_{i \neq j} E(X_i X_j) - E(X)^2$

- 0.5 pts $\sum_{i=1}^n Var(X_i) = 1 - \frac{1}{n}$, or $\sum_{i=1}^n E(X_i^2) = 1$

- 1 pt $\sum_{i \neq j} Cov(X_i, X_j) = 1 - \frac{n-1}{n}$, or $\sum_{i \neq j} E(X_i X_j) = 1$

- 0.5 pts $SD(X) = 1$

Click here to replace this description.

✓ - 0 pts Correct

- 4 pts Incorrect

- 0.5 pts Calculation mistake

$$E[X^2] - (E[X])^2$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Homework 6

$$1.1) \text{Var}(X|Y=0)$$

$$\text{Var}(X|Y=0) = E[X^2|Y=0] - (E[X|Y=0])^2$$

$X=1$	$\frac{3}{16}$	$\frac{1}{16}$	$\frac{1}{8}$
$X=0$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{4}$
	$Y=-1$	$Y=0$	$Y=1$

• From the table:

$$P(X=1, Y=0) = \frac{1}{16}$$

$$P(X=0, Y=0) = \frac{1}{8}$$

$$P(Y=0) = \frac{1}{16} + \frac{1}{8} = \frac{3}{16}$$

• Conditionals:

$$P(X=1|Y=0) = \frac{P(X=1, Y=0)}{P(Y=0)} = \frac{\frac{1}{16}}{\frac{3}{16}} = \frac{1}{3}$$

$$P(X=0|Y=0) = \frac{P(X=0, Y=0)}{P(Y=0)} = \frac{\frac{1}{8}}{\frac{3}{16}} = \frac{2}{3}$$

$$* E[X|Y=0] = \sum_X X \cdot P_{X|Y}(X|Y=0) = 0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \frac{1}{3}$$

$$* E[X^2|Y=0] = \sum_X X^2 \cdot P_{X|Y}(X^2|Y=0) = 0^2 \cdot \frac{2}{3} + 1^2 \cdot \frac{1}{3} = \frac{1}{3}$$

$$\text{Var}(X|Y=0) = \frac{1}{3} - \left(\frac{1}{3}\right)^2 = \frac{3}{9} - \frac{1}{9} = \boxed{\frac{2}{9}}$$

$$1.2) P_{X|Y}(1|A), \text{ where } A = \{Y^2 > 0\}$$

• $Y^2 > 0 \rightarrow Y \neq 0$, but can equal -1 or 1

• From the table:

$$P(X=1, Y=-1) = \frac{3}{16}$$

$$P(X=1, Y=1) = \frac{1}{8}$$

$$P(X=0, Y=-1) = \frac{1}{4}$$

$$P(X=0, Y=1) = \frac{1}{4}$$

$$\rightarrow P(Y \neq 0) = \frac{3}{16} + \frac{1}{8} + \frac{1}{4} + \frac{1}{4} = \frac{13}{16}$$

$$P(X=1|Y \neq 0) = \frac{P(X=1, Y \neq 0)}{P(Y \neq 0)} = \frac{\frac{3}{16} + \frac{1}{8}}{\frac{13}{16}} = \frac{\frac{5}{8}}{\frac{13}{16}} = \boxed{\frac{5}{13}}$$

HW 6 Con.

pdf → continuous

2.

$$f(x, y) = \begin{cases} \frac{16y}{x^3} & x > 2, 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

Verifying pdf validity: $\int_{x=2}^{\infty} \int_{y=0}^1 \frac{16y}{x^3} dy dx = \int_{x=2}^{\infty} \left(\int_{y=0}^1 16y dy \right) \frac{1}{x^3} dx$

Inner: $\int_0^1 16y dy = 16 \left[\frac{y^2}{2} \right]_0^1 = 8$

outer: $\int_2^{\infty} \frac{8}{x^3} dx = 8 \left[\frac{x^{-2}}{-2} \right]_2^{\infty} = 8 \left(0 - \left(-\frac{1}{8} \right) \right) = 1$

(i) $E[X] = \int_{x=2}^{\infty} \int_{y=0}^1 x \cdot \frac{16y}{x^3} dy dx = \int_{x=2}^{\infty} \int_{y=0}^1 \frac{16y}{x^2} dy dx$

$$\int_0^1 16y dy = 8 \quad \int_2^{\infty} \frac{1}{x^2} dx = 0 - \left(-\frac{1}{2} \right) = \frac{1}{2}$$

$\hookrightarrow 8 \cdot \frac{1}{2} = \boxed{4}$

$$E[Y] = \int_2^{\infty} \int_0^1 y \cdot \frac{16y}{x^3} dy dx = \int_2^{\infty} \int_0^1 \frac{16y^2}{x^3} dy dx$$

$$\int_0^1 16y^2 dy = \frac{16}{3} \quad \int_2^{\infty} \frac{1}{x^3} dx = 0 - \left(-\frac{1}{8} \right) = \frac{1}{8}$$

$$\frac{16}{3} \cdot \frac{1}{8} = \frac{16}{24} = \boxed{\frac{2}{3}}$$

(ii) $\text{Cov}(X, Y) = E[XY] - E[X] \cdot E[Y]$

$$E[XY] = \int_2^{\infty} \int_0^1 xy \cdot \frac{16y}{x^3} dy dx = \int_2^{\infty} \int_0^1 \frac{16y^2}{x^2} dy dx$$

$$\int_0^1 16y^2 dy = \frac{16}{3} \quad \int_2^{\infty} \frac{1}{x^2} dx = \frac{1}{2}$$

$$\frac{16}{3} \cdot \frac{1}{2} = \frac{16}{6} = \frac{8}{3}, \text{Cov}(X, Y) = \frac{8}{3} - 4 \cdot \frac{2}{3} = \boxed{0}$$

(iii)

No, X and Y are not correlated because $\text{Cov}(X, Y) = 0$.

If their covariance is zero, then the variables are uncorrelated.

Hw 6

3. • Catch type (Sarah) = p_1 • Miss type (Sarah) = $1-p_1$ • Catch type (Hannah) = p_2 • Miss type (Hannah) = $1-p_2$ X_1 : # of
types
caught (Sarah) X_2 : # of types
caught (Hannah)• $X^c \rightarrow$ # of types missed by both of them• $X \rightarrow$ # of types caught by at least one of them

$$\hookrightarrow P(X) = 1 - P(X^c) = 1 - (1-p_1)(1-p_2) = 1 - (1-p_2 - p_1 + p_1 p_2) \\ = p_2 + p_1 - p_1 p_2$$

a)

$$X \sim \text{Binomial}(n, 1 - (1-p_1)(1-p_2))$$

use for b: $\rightarrow X_1 \sim (n, p), X_2 \sim (n, p) \rightarrow X_1 + X_2 \sim (2n, p)$ * b) $p_1 = p_2 = p, X_1 + X_2 = t \rightarrow X_1$ and X_2 are also binomial r.v.s

$$P(X_1 = x | X_1 + X_2 = t) = \frac{P(X_1 = x, X_1 + X_2 = t)}{P(X_1 + X_2 = t)} = \frac{P(X_1 = x, X_2 = t - x)}{P(X_1 + X_2 = t)}$$

$$= \frac{P(X_1 = x) P(X_2 = t - x)}{P(X_1 + X_2 = t)} = \frac{P(X_1 = x) P(X_2 = t - x)}{P(X_1 + X_2 = t)}$$

$$= \frac{\binom{n}{x} p^x (1-p)^{n-x} \binom{n}{t-x} p^{t-x} (1-p)^{n-(t-x)}}{\binom{2n}{t} p^t (1-p)^{2n-t}}$$

$$= \frac{\binom{n}{x} \binom{n}{t-x}}{\binom{2n}{t}}$$

$$X_1 | X_1 + X_2 = t \sim \text{Hypergeometric}(2n, n, t)$$

- 2n opportunities for types to be caught
- n types caught by Sarah
- t types caught by both

* HW 6

* 4. $X = \#$ of heads in 10 fair coin tosses

$$\begin{matrix} \uparrow 8 \\ X = Y + 2 \end{matrix}$$

$$X \sim \text{Binomial}(n=10, p=\frac{1}{2})$$

 $Y = \#$ of heads in remaining tosses $\rightarrow Y \sim (n=8, p=\frac{1}{2})$

$$\begin{aligned} a) P(X=x | \text{first 2 tosses are head}) &= \binom{8}{x-2} \left(\frac{1}{2}\right)^{x-2} \left(1-\frac{1}{2}\right)^{8-(x-2)} \\ &= \binom{8}{x-2} \left(\frac{1}{2}\right)^8, \quad X=2-10 \end{aligned}$$

$$P(X=x | \text{first 2 tosses are head}) = \begin{cases} \binom{8}{x-2} \left(\frac{1}{2}\right)^8 & \text{if } X=2, 3, \dots, 10 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} * b) P(X=x) &= \binom{10}{x} \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{10-x} \\ &= \binom{10}{x} \left(\frac{1}{2}\right)^{10} \end{aligned}$$

$$P(X=x | X \geq 2) = \frac{P(X=x)}{P(X \geq 2)} = \frac{\binom{10}{x} p^x (1-p)^{10-x}}{1 - \binom{10}{0} p^0 (1-p)^{10} - \binom{10}{1} p^1 (1-p)^9}$$

$$P(X \geq 2) = 1 - P(X=0) - P(X=1) \rightarrow 1 - \sum_{x=0}^1 \binom{10}{x} \left(\frac{1}{2}\right)^{10}$$

$$\rightarrow 1 - \binom{10}{0} \left(\frac{1}{2}\right)^{10} - \binom{10}{1} \left(\frac{1}{2}\right)^{10} = 1 - \frac{1}{1024} - \frac{10}{1024} = \frac{1013}{1024}$$

$$P(X=x | X \geq 2) = \frac{\binom{10}{x} \left(\frac{1}{2}\right)^{10}}{\frac{1013}{1024}} = \frac{\binom{10}{x}}{1013}, \quad X=2, 3, \dots, 10$$

$$P(X=x | X \geq 2) = \begin{cases} \frac{\binom{10}{x}}{1013} & \text{if } X=2, 3, \dots, 10 \\ 0 & \text{otherwise} \end{cases}$$

$$Z = X + Y \rightarrow Z = X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$$

$$\hookrightarrow Y = Z - X$$

$$P(Z=z) = \sum_{x=0}^z P(X=x) P(Y=z-x)$$

$$= \sum_{x=0}^z \left(\frac{e^{-\lambda_1} \lambda_1^x}{x!} \right) \left(\frac{e^{-\lambda_2} \lambda_2^{z-x}}{(z-x)!} \right)$$

$$= e^{-(\lambda_1 + \lambda_2)} \sum_{x=0}^z \frac{\lambda_1^x \lambda_2^{z-x}}{x! (z-x)!}$$

Sum is a binomial expansion

$$= e^{-(\lambda_1 + \lambda_2)} \cdot \frac{(\lambda_1 + \lambda_2)^z}{z!}$$

$$= \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^z}{z!}$$

$$Z = X + Y \sim \text{Pois}(\lambda_1 + \lambda_2)$$

- Total # of people who draw their own name:

$$X = X_1 + X_2 + \dots + X_n$$

$$E[X] = \sum_{i=1}^n E[X_i] \rightarrow \text{Linearity of expectation}$$

slips uniformly random

- Each person has a $\frac{1}{n}$ chance of drawing their own name.

$$E[X_i] = P(X_i = 1) = \frac{1}{n} \quad E[X] = \sum_{i=1}^n \frac{1}{n} = 1$$

$$E[X] = n \cdot \frac{1}{n} = 1 \rightarrow \text{mean} \rightarrow \boxed{\text{one}} \quad E[X_i^2] = E[X_i] = \frac{1}{n}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 \quad \left\{ \begin{array}{l} E[X^2] = E[X_i^2] + \sum_{i \neq j} E[X_i X_j] \end{array} \right.$$

For $i \neq j$, need $E[X_i X_j] = P(\text{both } i \text{ and } j \text{ draw own names}) \rightarrow X_i X_j = 1$

$$\hookrightarrow P(X_i = 1 \cap X_j = 1) = P(X_i = 1) \cdot P(X_j = 1 | X_i = 1) = \frac{1}{n} \cdot \frac{1}{n-1} = \frac{1}{n(n-1)}$$

$$E[X^2] = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 1 + 1 = 2, \quad \text{Var}(X) = 2 - (1)^2 = 1 \rightarrow \boxed{\text{SD}(X) = 1}$$

